

BINAY - Lighting Up The Future



BINAY LED-based HIGH, MEDIUM and LOW Intensity Aviation Obstruction Light Beacons

As per International Civil Aviation Organisation (ICAO) requirements Available in Low Intensity, Medium Intensity and High Intensity versions (as per International Civil Aviation Organization guidelines), BINAY's patented LED Aviation Lights come with 5-year/3-year warranties

LED Obstruction Lighting for:

- Industrial chimneys and smokestacks
- Transmission, microwave and cellular towers
- Radio, TV and similar structural towers
- High-rise buildings and structures
- Airports and airfields

The BINAY LED Aviation Obstruction Light offers the following advantages:

- Fit-and-forget maintenance-free operation
- A long life of 100,000 hours (20 years at 12 hours daily burning)
- Pays for itself within a short period of operation in the form of reduced installation, maintenance and servicing costs
- Quick Installation; Reliable operation 365 days per year
- Shock-proof and vibration-resistant
- Over-Designed Intensity to allow for natural LED Intensity degradation over its operating lifetime



THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT

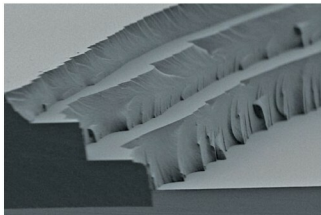
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POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983

Polymer super suit designed using solar energy

The Grossman group of MIT has developed a transparent polymer that can store energy by using a solar cell and release controllable heat at any time. This newly-engineered material depends upon the Sun, which is a practically inexhaustible source of energy and stores energy in the form of chemical energy, releasing it later as heat.



The layer-by-layer solar thermal fuel polymer film comprises three distinct layers (Image courtesy: www.news.mit.edu)

Jeffrey Grossman, lead researcher, has said that the product could be a boon for the clothing industry, and provide humans with a new type of protective wear. The team also explained that this concept first came to their mind while analysing the concept of harvesting solar energy for long-term usage, as required in various sectors.

Conventionally, solar energy is converted to electrical energy and serves as an environment-friendly renewable energy source, but the researchers wanted to come up with something new and innovative by using similar ideas.

Designer crystals for next-gen electronics

Liquid is often seen as the kryptonite of electronics, known for damaging and corroding components. This is why a new process that uses vapour instead of liquid to grow designer crystals could lead to a new breed of faster, more powerful electronic devices.

The method is invented by an international team of scientists from University of Leuven in Belgium, National University of Singapore and CSIRO. For the first time, researchers have shown how designer crystals known as metal organic frameworks can be grown using a vapour method that is similar to steam hovering over a pot of hot water.

The crystals are the world's most porous materials, and if applied to microelectronic devices, could significantly boost their processing power.